

Phase D2 Third-Quarter Synthesis and Final Quarter Preview

Series D — Paper 3 Derivation Note D2.60 (Note #555)

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This work derives from and formalizes the inter-Self coordination architecture introduced in "Inter-Self Coordination via Shared Substrate / Full Aspect Integration" (Li, April 2026), the third paper in the CKS theory series following "Coordination Outside the Model: A Human-Governed Substrate Content for AI Systems" (Li, April 2026) and "The Instinct/Reasoning Separation Outside the Model" (Li, April 2026).

Abstract

D2.60 is the third-quarter synthesis note for Phase D2 of the CKS derivation note series, marking completion of D2.01 through D2.59 — three quarters of Phase D2's approximately eighty operational-variant notes for Paper 3. The note summarizes what the third quarter (D2.51–D2.59) added across four categories: inter-claim operational synthesis establishing configuration governance over conflict handling and evolution feed; cross-paper lifecycle and evolution integration establishing the lifecycle isolation principle; Self-level and vertical evolution extending FAI learning through the full three-tier governance hierarchy; and multi-domain and proposing-substrate governance completing the home-governance cycle. The note then defines what the final quarter (D2.61–D2.80) will complete: population-scope operations for Claim 6, governance operational completeness on the remaining edge cases, targeted cross-paper synthesis notes, and the Phase D2 closure note at D2.80. A coverage-completeness assessment by claim identifies Claims 1–5 as comprehensively covered and Claim 6 as partially covered, with the specific Claim 6 sub-commitments requiring additional notes identified. The note concludes that Phase D2's core prior-art function is substantively complete at D2.60 — the operational prior art for Paper 3 is established; the final twenty notes add depth, completeness, and closure rather than fundamentally new territory.

1. Position of D2.60 in Phase D2

Phase D2 derives operational variants from Paper 3's six architectural claims, following the same structure that Phases A2 and B2 applied to Papers 1 and 2 respectively. Each D2 note takes a sub-commitment established in Phase D1 and formalizes the operational specifics, edge cases, and governing conditions that a practitioner or examiner would need to evaluate whether a given system instantiates the commitment. Phase D2 spans approximately eighty notes (D2.01–D2.80), placing D2.60 at the three-quarter mark.

D2.60 is a synthesis note. It does not introduce a new operational variant. Its function is to consolidate what the third quarter established, assess the current coverage state by claim, define the scope of the remaining work with enough specificity that the final quarter can proceed without further scoping, and formally note that Phase D2's core prior-art contribution is achieved.

Three prior synthesis notes structure Phase D2's progress: D2.20 (first-quarter synthesis) marked the establishment of foundational per-claim operational coverage for Claims 1–4; D2.40 (second-quarter synthesis) marked the extension of coverage to Claim 5 and the minimum viable governance floor; and D2.60 (this note) marks the completion of inter-claim synthesis, lifecycle integration, and the governance cycle reaching from FAI events back through home-substrate improvement. D2.80 will mark final closure and transition to Phase D3.

2. What the Third Quarter Added (D2.51–D2.59)

2.1 Inter-Claim Operational Synthesis (D2.51–D2.52)

The third quarter opened by closing the most significant gap that remained after D2.50: the adversarial gap between Claim 5's configuration-as-substrate-content commitment and the operational mechanics of Claims 3 and 4. D2.51 established that conflict handling — the three-tier preserve/resolve/escalate mechanism of Claim 3 — is a governed, configured system whose tier-selection logic, escalation thresholds, and resolution rules are themselves substrate content subject to all of Claim 5's recursive applicability. D2.52 established the same for the evolution feed: the four-locus mechanism of Claim 4, including the layer-routing rule, the per-mechanism feed structure, and the asymmetric ingestion policy, is governed and configured through substrate content rather than being a fixed implementation choice baked below governance reach.

These two notes are the inter-claim adversarial closure notes for Phase D2. An adversary could challenge Paper 3 by arguing that while Claims 3 and 4 describe the conflict-handling and evolution-feed mechanisms, Claim 5's governance commitment does not actually reach those mechanisms — that configuration governs the "what" of inter-Self coordination but not the "how" of its internal dynamics. D2.51 and D2.52 establish the prior art that closes that gap operationally: the mechanisms are governed systems, not fixed systems, and the configuration governing them is substrate content subject to the same recursive applicability Claim 5 commits to across all dimensions.

2.2 Cross-Paper Lifecycle and Evolution Integration (D2.53–D2.55)

D2.53 through D2.55 addressed a distinct adversarial exposure: the interaction between FAI operations (Paper 3 scope) and the lifecycle and evolution machinery those FAI events feed into (Paper 2 scope). The risk is that race conditions or interference between home-lifecycle events and FAI operations could undermine the clean governance boundaries Paper 3 requires.

D2.53 established the FAI/lifecycle interaction principle: home lifecycle events — birth, maturity, transformation, dissolution — are isolated from FAI operations except through explicit governance action. An FAI event does not implicitly trigger home lifecycle transitions, and a home lifecycle event during an active FAI operation does not automatically terminate or corrupt that operation. The version-fixing principle

is the operational mechanism: a Self's contributed aspects are fixed at contribution time, so home governance events occurring during the FAI event do not retroactively alter what was contributed. D2.54 extended this to the evolution feed: FAI dissolution events hand off to the home evolution machinery, but the hand-off is governed and bounded — the evolution mechanisms receive what the FAI event produced under the layer-routing rule, not a direct write into the Self's ongoing lifecycle state.

D2.55 established aspect restructuring as the mechanism through which FAI learning enters the home substrate in restructured form rather than as raw FAI content appended without governance processing. Together, D2.53–D2.55 establish the lifecycle isolation principle as an operational commitment: home lifecycle and evolution events are isolated from FAI operations except through explicit governance action, and the governance actions that do bridge the two scopes are themselves substrate content subject to Claim 5.

2.3 Self-Level and Vertical Evolution (D2.56–D2.57)

D2.56 and D2.57 extended the evolution integration to the highest governance tier within the three-tier hierarchy. D2.56 established how the Self architecture — the tier at which governance rules governing all lower cells are maintained — evolves through accumulated FAI learning. FAI events produce learning that, through the aspect restructuring and evolution-feed mechanisms D2.53–D2.55 formalized, eventually reaches the Self's governance configuration. This is not automatic: the escalation tier of Claim 3's conflict-handling mechanism is the governance pathway through which inter-Self learning that cannot be resolved at the orchestration tier reaches the humans responsible for Self-architecture governance.

D2.57 established vertical evolution as the mechanism through which FAI learning propagates through the full three-tier hierarchy — from the cell-level action-layer content where individual task instances are recorded, through DNA-layer content where behavioral patterns and schemas are maintained, to the Self-architecture level where the governance rules themselves may be updated. The three-case layer-routing rule of Claim 4 is the operational spine of vertical evolution: DNA-layer content routes to DNA evolution, action-layer content routes to action-feedback evolution, and instinct evolution takes no FAI input by architectural commitment. D2.57 formalized how these routing constraints govern the directionality and reach of vertical evolution — ensuring that FAI learning amplifies the capacity for governance-configured evolution without collapsing the instinct/reasoning separation Paper 2 establishes.

2.4 Multi-Domain Governance and the Proposing Substrate (D2.58–D2.59)

D2.58 addressed domain-selective governance: a Self that operates across multiple functional domains may govern FAI events selectively by domain, contributing aspects from some domains while withholding aspects from others, and applying different evolution-feed governance rules by domain. The domain-selective governance commitment is not a relaxation of Claim 5 — it is an application of Claim 5's recursive applicability to the dimension of domain scope. The configuration of which domains participate in a given FAI event, and under what governance rules, is itself substrate content.

D2.59 completed the governance cycle by establishing the proposing-substrate interaction: the home governance mechanism through which a Self's governance authorities propose updates to the Self's

architecture, behavioral rules, or aspect configurations based on accumulated FAI learning. The proposing substrate is the substrate-content record of proposed updates awaiting human governance review, distinct from the substrate-content record of enacted updates. D2.59 formalized this as the closing link in the governance cycle: FAI events produce learning that enters the home substrate through the evolution-feed mechanism, accumulates in the proposing substrate as candidate improvements, and reaches governance authorities for review and potential enactment — closing the cycle from inter-Self coordination back to home-governance improvement.

3. What the Final Quarter Will Complete (D2.61–D2.80)

The final twenty notes of Phase D2 have four defined categories. Each is described here with enough specificity to allow prompt-level execution without additional scoping.

3.1 Population-Scope Operations for Claim 6 (D2.61–D2.68, approximately)

Claim 6 articulates population-scale collective evolution as what Claims 1–5 compose to at population register when accumulated FAI events across many CKS-governed Selves operate under joint authority across population-level governance perimeters. Phase D2's coverage of Claim 6 through D2.59 is partial: D2.31–D2.32 addressed population-scope basics and D2.50's preview identified the remaining gaps. The final quarter must add operational prior art for three specific Claim 6 sub-commitments:

FAI network topology governance.

How the joint-authority architecture governing a network of CKS-governed Selves is structured, configured, and maintained as substrate content. A network topology is not a fixed graph — it is a governed configuration specifying which Selves can engage in FAI events with which others, under what conditions, and under which participating Selves' joint governance. The operational prior art must formalize how topology governance is itself substrate content with recursive applicability.

Network-level pattern propagation control.

How patterns derived from accumulated FAI events — learned behavioral improvements, successful aspect configurations, evolved governance rules — propagate across the network under joint authority, rather than remaining siloed within individual Selves. Propagation control is the governed mechanism that prevents ungoverned pattern spread while enabling the collective-evolution benefit Claim 6 targets.

Population-scale governance volume mechanics.

How the joint-authority architecture scales to many-Self, high-volume FAI event accumulation without either collapsing governance to unilateral control by one Self's governance authority or dissolving into ungoverned collective dynamics. The minimum viable governance floor D2.37 established for individual FAI operations requires a population-scope analog that the final quarter formalizes.

3.2 Governance Operational Completeness (D2.69–D2.73, approximately)

A small set of governance operational topics, not directly tied to Claim 6 population scope, remain uncovered after D2.59. These include FAI event recording for historical research contexts, where

governance obligations apply when FAI events involve content designated as historically significant and where the evolution-feed mechanism's standard dissolution handling may need to be superseded by preservation obligations. A second topic is governance for regulatory audit purposes — how the governance-configured-persistence commitment of Claim 1 is applied when regulatory audit requirements specify retention and inspection obligations exceeding the standard minimum viable governance floor, formalized as an application of Claim 5's recursive applicability rather than as an exception. A third topic covers edge cases in the minimum viable governance floor: governance for ultra-short-duration FAI events where dissolution occurs before standard recording completes; governance when one participating Self's governance authority becomes unavailable mid-event; and governance for FAI events producing no aspects routing to any evolution locus.

3.3 Cross-Paper Synthesis Notes (D2.74–D2.77, approximately)

D2.53–D2.59 addressed the key Paper 2 / Paper 3 interaction points in the context of specific operational variants. The final quarter adds synthesis notes that explicitly trace how Phase D2's operational prior art closes specific adversarial claims that might target the Paper 2 → Paper 3 transition — arguments that Paper 3's mechanisms are not genuinely distinct from Paper 2's intra-Self mechanisms, or that the inter-Self perimeter is merely a relabeling of Paper 2 concepts at larger scope. These notes do not introduce new operational variants. They collect and cross-reference the relevant D2 notes and articulate the adversarial gap each closes, providing a structured reference for the cross-derivation work that Series CC will formalize.

3.4 Phase D2 Final Closure (D2.80)

D2.80 is the Phase D2 closure note. It consolidates what Phase D2's approximately eighty notes established, maps the full prior-art coverage by claim, identifies the adversarial gaps that Phase D2 closes and those whose closure is completed in subsequent phases, and transitions to Phase D3 (anti-pattern formalizations). D2.80 is the governance artifact that makes the transition to D3 a documented, justified decision rather than a sequence boundary.

4. Coverage Completeness Assessment by Claim

This assessment is the key governance artifact of D2.60. It states, by claim, what Phase D2 has established and what remains.

Claim 1 — Shared substrate as the architectural object: COMPREHENSIVE.

D2.01–D2.10 established the full prior-art record for Claim 1's sub-commitments: temporary construction and dissolution, perimeter-spanning property, governance-configured persistence, all six Paper 1 commitments holding within scope, and the configuration of the perimeter-spanning itself as substrate content under joint human authority. No material Claim 1 operational gaps remain after D2.59.

Claim 2 — Full Aspect Integration as the canonical operation: COMPREHENSIVE.

D2.11–D2.22 established prior art for FAI's sub-commitments: aspect as exchange unit, n-ary cardinality, full-merge architectural default, three pattern variants under orchestration-configured selection, exchange bounded to reasoning-layer content, and three persistence loci. D2.51's inter-claim synthesis added the governed-mechanism treatment of FAI's configurable dimensions. No material Claim 2 operational gaps remain after D2.59.

Claim 3 — Three-tier conflict-handling mechanism: COMPREHENSIVE.

D2.23–D2.31 established prior art for the preserve, resolve, and escalate tiers, with cross-perimeter escalation mechanics addressed in depth. D2.51 closed the inter-claim gap by establishing conflict handling as a governed, configured system. No material Claim 3 operational gaps remain after D2.59.

Claim 4 — Four-locus evolution-feed mechanism: COMPREHENSIVE.

D2.32–D2.42 established prior art for the dissolution hand-off, per-mechanism feed structure, layer-routing rule, instinct-takes-no-FAI-input commitment, and asymmetric ingestion. D2.52 closed the inter-claim gap by establishing the evolution feed as a governed, configured system. D2.53–D2.55 established lifecycle isolation. D2.56–D2.57 established vertical evolution mechanics. No material Claim 4 operational gaps remain after D2.59.

Claim 5 — Configuration as substrate content with recursive applicability: COMPREHENSIVE.

D2.43–D2.52 established prior art for all configurable dimensions as substrate content, configuration-of-configuration as substrate content, recursive applicability, and recursion bottoming at human-authored governance authority. The inter-claim synthesis notes D2.51 and D2.52 demonstrated recursive applicability against Claims 3 and 4 operationally. No material Claim 5 operational gaps remain after D2.59.

Claim 6 — Population-scale collective evolution: PARTIAL.

D2.31–D2.32 addressed population-scope basics at the level required for those notes' primary claims. D2.50 previewed the remaining Claim 6 operational territory. The three specific sub-commitments identified in §3.1 — FAI network topology governance, network-level pattern propagation control, and population-scale governance volume mechanics — require the dedicated operational treatment the final quarter will provide. The prior-art foundation for Claim 6 is established; the final quarter adds the depth that Claim 6's status as an extension claim requires.

5. The Core Prior-Art Function Is Complete

The primary strategic purpose of Phase D2 is to establish operational prior art for Paper 3's Claims 1–5 across all major sub-commitments. That purpose is achieved at D2.60.

An examiner evaluating Paper 3's Claims 1–5 for novelty or non-obviousness after D2.60 encounters a comprehensive record of operational variants, governance conditions, compliance tests, and inter-claim synthesis that covers every major sub-commitment. The minimum viable governance floor (D2.37) and the operational test suite (D2.24) provide the compliance verification infrastructure. The lifecycle isolation principle (D2.53–D2.55) and vertical evolution mechanics (D2.56–D2.57) cover the cross-paper interaction territory. The inter-claim synthesis notes (D2.51–D2.52) close the adversarial gaps between Claim 5 and the operational mechanisms of Claims 3 and 4.

This does not mean the final twenty notes are cosmetic. Claim 6's partial coverage is a genuine gap — population-scale collective evolution is the claim that positions Paper 3 at the largest governance scope, and the three sub-commitments identified in §3.1 require dedicated prior art. The governance operational completeness notes close real edge cases that could otherwise become adversarial openings. The cross-paper synthesis notes provide the organized reference material that Series CC will require. And D2.80's closure function is genuinely necessary — it is the documented transition rather than a mere sequence boundary.

The characterization is precise: the final quarter completes and deepens a prior-art record whose core contribution is already established. D2.61–D2.80 add depth, Claim 6 completeness, and clean closure. They do not establish the foundation — that work is done.

6. Transition to Phase D3

Phase D3 will formalize anti-patterns for Paper 3's commitments, following the structure Phases A3 and B3 applied to Papers 1 and 2. The expected categories map to the six claims: opaque inter-Self communication as the Claim 1 anti-pattern; shallow merge or federation-only as the Claim 2 anti-pattern; silent conflict resolution at the inter-Self tier as the Claim 3 anti-pattern; ungoverned evolution feed as the Claim 4 anti-pattern; fixed configuration or meta-governance escape as the Claim 5 anti-pattern; and emergent coordination replacing the substrate as the Claim 6 anti-pattern.

Phase D3 depends on Phase D2's operational coverage in the same way Phase B3 depended on Phase B2: each anti-pattern note names a failure mode with reference to the operational commitments it violates. Phase D2's comprehensive Claims 1–5 coverage, and its near-complete Claim 6 coverage at the close of D2.80, provides the operational foundation Phase D3 requires to be precise rather than merely nominal in its anti-pattern formalization.

The transition from D2 to D3 is a governance decision that D2.80 will formalize. The present note states that the prerequisite condition — substantive operational prior art for all six Paper 3 claims — will be met at D2.80, and that Phase D3 may proceed immediately after.

Note D2.60 of the CKS Series D derivation notes. Phase D2 third-quarter synthesis marking completion of D2.01–D2.59. Transition to Phase D2 final quarter (D2.61–D2.80) follows.